WILDTRAIL

Safari Booking System

ONE WEB APPLICATION · SIX CONNECTED MODULES

SE1020 • Object Oriented Programming

Group 26-MTR-Y1S2-WD-009



Six contributors, one project

#	Student	Member	Component	Main demonstration
01	IT26100894	Mekhala B.G.R.S	Users and customers	Register, profile, staff access
02	IT26100895	Mihiran S.P.D	Safaris and packages	Catalogue, price, package CRUD
03	IT26100896	Nayanajith W.R.S.C	Bookings	Availability, change, cancel
04	IT26100899	Senevirathna M.K.K.T	Guides and vehicles	Schedules, guide file CRUD
05	IT26100900	Lakshani W.G.C	Payments	Pending, cash, receipt, refund
06	IT26100901	Gamage C.I.P	Reviews	Submit, moderate, edit, delete

Each row leads to its classes, pages and tests in README → Complete project map.

One customer journey crosses all six modules

01 Customer signs in → 02 Chooses a safari → 03 Books a date →

04 Gets vehicle/guide → 05 Pays on arrival → 06 Leaves a review ✓

Controller → service/model → repository. Business rules live in Java, not only in the browser.

Java classes connected across the application

01 User

User → Customer / Admin
AuthService

02 SafariPackage

Seven safari subtypes
PackageService

04 Vehicle / Guide

Jeep subtypes · NaturalistGuide
VehicleService

03 Booking

BookingService · Booking
JdbcBookingRepository

05 Payment

PaymentProcessor → Payment
CashPayment / CardPayment

06 Review

Review → VerifiedReview
ReviewService

Every component has a clear data path

Component	Create / Read / Update / Delete example	Primary store
Users	Register · profile · remove eligible account	H2 SQL
Safaris	Add · filter · edit · remove unused package	H2 SQL
Bookings	Reserve · history · change · cancel	H2 SQL
Guides / vehicles	Add · schedule · edit · remove unused	Guide file + H2 SQL
Payments	Pending · receipt · status · failed removal	H2 SQL
Reviews	Submit · view · edit · moderate/delete	H2 SQL

Guide file: resources.txt • Payment card option: simulation only; no card data or charge.

Four concepts visible in the Java code

Encapsulation

Booking guards status, dates and price.

Inheritance

SafariPackage → concrete safari types.

Polymorphism

calculatePrice() varies by subtype.

Abstraction

PaymentProcessor defines processing.

A viva answer should name a class, a method and the behavior it changes.

A single demo through the whole application

01 Browse packages and destinations

02 Register and create a booking

03 Show an eligible change or cancellation

04 Inspect guide / vehicle schedule

05 Record cash collection and receipt

06 Submit and moderate a review

24-hour notice and availability checks are enforced by the Java API.

Tests, Git history and project documents

57
Java tests

11
Browser flows

4
Access checks

12
Viewport widths

Individual review trail

#14 Mekhala #12 Mihiran #7/#9 Nayanajith
#16 Senevirathna #10 Lakshani #11 Gamage

Six short answers to remember

Where are rules?

Java service and model.

Where is data?

H2 SQL; guides in resources.txt.

How is staff access checked?

Server-side session and role.

How are booking conflicts stopped?

Transactions, locks and validation.

Does card payment charge?

No. Demo only, pending.

Why 24 hours?

Notice protects planned resources.